

# Quantitative Investing

## PROBLEM SET 3 — PART I AND PART III, EXERCISE 2

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### 1 PART I: 49 INDUSTRY PORTFOLIOS

**Data and conventions.** The 49 value-weighted industry portfolios, their average firm size and book-to-market ratios, and the market proxy are taken from `Problem_Set3_2026.xlsx`. All figures are in percent per month. Missing observations are screened at  $-99$ , which removes the  $-99.99$  and  $-999$  codes; the assignment’s gloss of “any return  $< -1$ ” assumes decimal returns, and applied literally to percent data would discard 19,602 valid observations against 2,868 genuine missing values. Industry returns are total returns while the market proxy is reported as  $R_M - R_f$ , so we subtract the monthly riskless rate from each industry return and work in excess returns throughout. Six industries have no data before the late 1960s, and all 49 series are populated only from July 1969; we therefore report the balanced July 1969 – June 2026 panel ( $T = 684$ ,  $N = 49$ ) as our primary specification and the full July 1926 – June 2026 sample ( $T = 1200$ ,  $N$  from 40 to 49) as a robustness check. Eleven negative book-equity observations are masked, since the logarithm is undefined for them. The sample mean of  $R_M - R_f$  is 0.6116 on the balanced sample and 0.6952 on the full sample.

#### 1.1 Question (a): CAPM predictions for $\gamma_0$ and $\gamma_M$

In the cross-sectional regression  $R_i = \gamma_0 + \gamma_M \beta_{iM} + \eta_i$  with  $\beta_{iM} = \text{cov}(R_i, R_M) / \sigma^2(R_M)$ , each observation is a portfolio, so  $\gamma_0$  is the expected return on a portfolio with zero market beta and  $\gamma_M$  is the expected return per unit of beta.

**Sharpe/Lintner.** With unrestricted borrowing and lending at the riskless rate,  $E[R_i] = R_f + \beta_{iM}(E[R_M] - R_f)$ , so

$$\gamma_0 = R_f, \quad \gamma_M = E[R_M] - R_f.$$

Both are exact point predictions. A portfolio with  $\beta_{iM} = 0$  bears no systematic risk and must earn the riskless rate, which fixes the intercept. The market has  $\beta_{iM} = 1$ , so the line must pass through  $(1, E[R_M])$ , which fixes the slope at the market risk premium.

**Black.** Without riskless borrowing and lending,  $R_f$  is replaced by the expected return on the zero-beta portfolio  $R_z$ , the minimum-variance portfolio uncorrelated with the market, giving  $E[R_i] = E[R_z] + \beta_{iM}(E[R_M] - E[R_z])$  and

$$\gamma_0 = E[R_z], \quad \gamma_M = E[R_M] - E[R_z].$$

Because  $E[R_z]$  is not observable, this version predicts no specific intercept. It restricts only  $\gamma_M > 0$  and  $\gamma_0 < E[R_M]$ , and is therefore weaker and harder to reject: Sharpe/Lintner is Black plus the restriction  $E[R_z] = R_f$ . Both imply  $\gamma_0 + \gamma_M = E[R_M]$ , so they differ only in where the security market line is anchored.

Because all tests use excess returns, the Sharpe/Lintner null becomes  $\gamma_0 = 0$  with  $\gamma_M$  equal to the average market excess return, and Black predicts  $\gamma_0 = E[R_z] - R_f$ , unrestricted in sign but generally positive.

## 1.2 Question (b): Fama-MacBeth estimation

Each industry's beta is estimated once by OLS over the full sample and held fixed thereafter; the estimates range from 0.516 (Utilities) to 1.514 (Software), with mean 1.039. In each of the 684 months we then estimate  $R_{it} = \gamma_{0t} + \gamma_{Mt}b_{iM} + \eta_{it}$  across the 49 industries. The monthly slopes are highly volatile, with a standard deviation of 7.13 against a mean of 0.035, and are positive in only 48.4% of months. Finally we average the monthly estimates, using

$$\hat{\gamma}_j = \frac{1}{T} \sum_{t=1}^T \hat{\gamma}_{jt}, \quad s(\hat{\gamma}_j) = \frac{\sigma(\hat{\gamma}_{jt})}{\sqrt{T}},$$

so that the standard error is the time-series standard deviation of the monthly estimates divided by  $\sqrt{T}$ , not the cross-sectional OLS standard error from any single month.

**Table 1:** Fama-MacBeth estimates, 49 industry portfolios, percent per month. The final column gives the value each coefficient should take if the CAPM holds, with  $\gamma_0 = 0$  under Sharpe/Lintner in excess-return space.

|                                                              | Estimate | Std. Error | $t$  | $p$   | CAPM   |
|--------------------------------------------------------------|----------|------------|------|-------|--------|
| <i>Balanced, July 1969 – June 2026, <math>T = 684</math></i> |          |            |      |       |        |
| $\gamma_0$                                                   | 0.6119   | 0.2191     | 2.79 | 0.005 | 0      |
| $\gamma_M$                                                   | 0.0352   | 0.2725     | 0.13 | 0.897 | 0.6116 |
| <i>Full, July 1926 – June 2026, <math>T = 1200</math></i>    |          |            |      |       |        |
| $\gamma_0$                                                   | 0.5160   | 0.1714     | 3.01 | 0.003 | 0      |
| $\gamma_M$                                                   | 0.2490   | 0.2315     | 1.08 | 0.282 | 0.6952 |

**Can we reject mean-variance efficiency?** Yes. Both restrictions fail: the intercept is significantly positive where it should be zero, and the slope is indistinguishable from zero and significantly below the realized market premium ( $t = -2.12$ ). Taken together these describe a single pattern, the flat security market line, which is too high at  $\beta = 0$  and too shallow thereafter. That pattern licenses the stronger conclusion, because the exact linear relation between  $E[R_i]$  and  $\beta_{iM}$  holds if and only if the proxy is mean-variance efficient with respect to the test assets. The positive intercept alone would be consistent with Black, which permits  $E[R_z] > R_f$ , but that reading requires the zero-beta rate to absorb essentially the entire market premium while beta goes unpriced. The full sample gives the same conclusion with slightly less extreme flattening.

**AI integration: code generation.** We first asked an LLM to explain three aspects of the procedure, limiting it to three lines each, so the omissions below reflect our prompt rather than its knowledge.

- **Fixed betas.** It answered that betas need time-series variation to estimate and that short rolling windows would let noise correlate mechanically with the same month's return. This matches our understanding. We would add that fixed betas also make the monthly  $\hat{\gamma}_{Mt}$  comparable across months.
- **Standard errors.** It correctly identified cross-sectional correlation in residuals as the source of the difference and predicted correctly that the Fama-MacBeth errors would be larger. It did not note that the two methods give identical point estimates in a balanced panel.
- **Errors-in-variables.** It correctly described attenuation of the slope and the offsetting upward bias in the intercept, without adding that Fama-MacBeth does not fix this, since the same estimated betas enter every month.

We then asked it to implement the procedure and ran its script ourselves.

- (i) The delivered code was correct; the errors it reported were from earlier drafts, namely a 36-month rolling beta, a missing  $\sqrt{T}$  divisor, and an assumed fixed cross-section of 49. Two corrections were ours: it ran the full rather than the balanced sample, and it described the missing codes as “−99.99 or −999” when only −99.99 occurs here.
- (ii) Yes. It divides the time-series standard deviation by  $\sqrt{T}$ , and our independent run reproduces its standard errors exactly. It contrasted this against a pooled panel OLS rather than against the cross-sectional regression the assignment describes.
- (iii) Yes on substance, but this assessment is the model’s own, so we treat it as a claim and verified it by re-running the script against our data.

### 1.3 Question (c): Cross-sectional regression on time-averaged returns

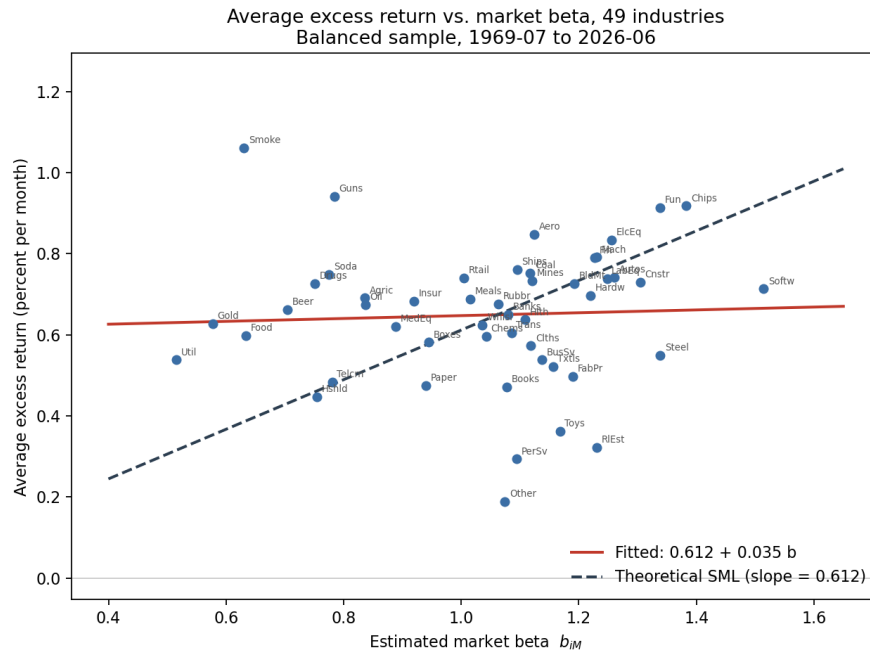
**Table 2:** Fama-MacBeth compared with OLS on time-averaged returns, balanced sample. The point estimates are identical; only the standard errors differ.

|                     | $\gamma_0$ |            | $\gamma_M$ |            |
|---------------------|------------|------------|------------|------------|
|                     | Estimate   | Std. Error | Estimate   | Std. Error |
| (b) Fama-MacBeth    | 0.6119     | 0.2191     | 0.0352     | 0.2725     |
| (c) OLS on averages | 0.6119     | 0.1154     | 0.0352     | 0.1086     |

**Are the estimates different?** No. Because the betas are fixed and the panel is balanced, every monthly regression uses the same regressor, so each monthly coefficient is a weighted average of that month’s returns with constant weights. Averaging those coefficients over time is the same calculation as regressing the time-averaged returns on beta. The estimates coincide exactly only in the balanced panel; in the full sample the number of industries varies across months, so the two methods weight months differently.

**Are the standard errors different, and which method is superior?** The Fama-MacBeth standard errors are roughly twice as large, because the two methods measure different sources of uncertainty. The cross-sectional regression treats the average returns as fixed and uses only the dispersion of the 49 residuals, which averaging over time has already made small. Fama-MacBeth instead uses the month-to-month variation in the estimated coefficients, which is large. Fama-MacBeth is superior: since the point estimates are the same, only the standard errors matter, and the cross-sectional regression assumes residuals are independent across industries. That assumption is not reasonable here, because industries share a common market shock. On the full sample this matters for the conclusion, as the cross-sectional regression makes  $\hat{\gamma}_M$  appear significant while Fama-MacBeth does not.

### 1.4 Question (d): Average return against estimated beta



**Figure 1:** Average excess return against estimated beta, 49 industry portfolios, July 1969 – June 2026. Solid line: fitted cross-sectional regression,  $\bar{R}_i = 0.612 + 0.035 b_{iM}$  with  $R^2 = 0.002$ . Dashed line: theoretical SML through the origin with slope 0.612.

**Does the plot resemble a positive relationship?** Barely. The fitted line is almost flat, and beta explains essentially none of the cross-sectional variation in average returns. Low-beta industries such as Utilities, Gold and Food earn about as much as high-beta industries such as Software, Chips and Steel. The vertical scatter at any given beta is much wider than the variation explained by beta itself, which is visible around  $b_{iM} \approx 1.1$ , where average returns range from Other near the bottom of the chart to Ships and Coal near the top.

**What should the plot look like?** Under the CAPM the points should lie along an upward-sloping straight line. In excess-return space that line passes through the origin, because a zero-beta portfolio should earn no premium, and its slope should equal the average market excess return. Deviations should be small and unsystematic, because beta is supposed to be the only characteristic that is priced. Instead the fitted line is far too high at  $\beta = 0$  and far too shallow, so it crosses the theoretical line near  $\beta = 1$  and diverges from it in both directions. This is the flat security market line of part (b) shown graphically.

**AI integration: SML prediction.** Asked what the scatter should look like and what the most common deviation is, the LLM answered that the points should lie on a line through the origin with slope equal to the market premium, that the estimated line is typically too flat with a positive intercept, and that this reflects leverage constraints under Black’s zero-beta CAPM, citing the Frazzini-Pedersen “betting against beta” result, and omitted risk factors such as size and value.

- (i) The empirical SML is much flatter than the theoretical one. The model predicted the direction correctly but understated the degree, describing a line that still slopes upward, whereas our slope is statistically indistinguishable from zero.

- (ii) It mentions all three, and its account of the leverage mechanism is correct. It does not mention errors-in-variables in the estimated betas, which produces the same flattening for purely statistical reasons, or Roll's critique, under which a flat line may indicate a poor market proxy rather than a failed model. The response also referred to our results, so it was not predicting blind.
- (iii) No. Beta explains almost none of the cross-sectional variation, and industries with very different betas earn similar average returns. Utilities and Software have the lowest and highest betas in the sample, yet their average returns differ by little.

### 1.5 Question (e): CAPM predictions for $\gamma_{size}$ and $\gamma_{B/M}$

In the extended regression

$$R_i = \gamma_0 + \gamma_M \beta_{iM} + \gamma_{size} \ln(\text{size}) + \gamma_{B/M} \ln(\text{BE/ME}) + \eta_i,$$

the CAPM predicts  $\gamma_{size} = \gamma_{B/M} = 0$ . Expected return depends only on covariance with the market, so beta is a sufficient statistic and no other characteristic should have explanatory power once beta is included. Size and book-to-market are firm characteristics rather than measures of systematic risk, so they should be priced only if they proxy for beta, and beta is already controlled for. The predictions for  $\gamma_0$  and  $\gamma_M$  are unchanged from part (a).

This regression tests the Fama-French (1992) critique. A significantly negative  $\gamma_{size}$  would indicate a size premium and a significantly positive  $\gamma_{B/M}$  a value premium. Either result rejects the CAPM, regardless of the estimate of  $\gamma_M$ .

### 1.6 Question (f): Extended Fama-MacBeth estimation

Using the same fixed betas, we estimate each month

$$R_{it} = \gamma_{0t} + \gamma_{Mt} b_{iM} + \gamma_{size,t} \ln(\text{size}_{t-1}) + \gamma_{B/M,t} \ln(\text{BE/ME}_{t-1}) + \eta_{it},$$

with size lagged one month and book-to-market taken from the previous year and held constant within the year.

**Why must the characteristics be lagged?** They must be known to investors before the return is realized. Size is measured by market capitalization, which moves with the return itself, so contemporaneous size would place part of the month's return on both sides of the regression and create a spurious relationship. Book-to-market has the same problem through its market value denominator, and annual accounting data is not public until several months after the fiscal year ends. Lagging makes the regression a test of whether information available at the start of the month predicts the return earned over it.

**Table 3:** Extended Fama-MacBeth estimates, balanced sample, July 1969 – June 2026 ( $T = 684$ ), percent per month. The average cross-sectional  $R^2$  rises from 0.09 in part (b) to 0.18.

|                 | Estimate | Std. Error | $t$  | CAPM   |
|-----------------|----------|------------|------|--------|
| $\gamma_0$      | 0.4045   | 0.4988     | 0.81 | 0      |
| $\gamma_M$      | 0.0328   | 0.2595     | 0.13 | 0.6116 |
| $\gamma_{size}$ | 0.0428   | 0.0497     | 0.86 | 0      |
| $\gamma_{B/M}$  | 0.1568   | 0.0805     | 1.95 | 0      |

**Can we reject mean-variance efficiency?** Yes, for the same reason as in part (b): the slope on beta is still indistinguishable from zero and significantly below the average market excess return ( $t = -2.23$ ). Adding the characteristics does not restore a risk-return relationship.

The characteristics themselves give only weak evidence against the CAPM. The coefficient on book-to-market is positive and marginally significant, which is consistent with a value premium, while the coefficient on size is small and insignificant. This is not surprising for industry portfolios, which are not sorted on these characteristics and therefore show limited cross-sectional variation in them; the 25 portfolios in Part II provide a sharper test. The intercept is no longer significant once the characteristics are included, which suggests the zero-beta return found in part (b) is partly absorbed by book-to-market rather than being a separate effect. Over the full sample both characteristics are insignificant and the intercept is significantly positive, with the rejection again coming from the slope on beta ( $t = -2.83$ ).

**AI integration: interpreting size and B/M premia.** The prompt specified in the assignment asserts significant premia on both characteristics, which our results do not show, so we ran it as written and then supplied our actual table, to test whether the model would confirm the premise it was given. Asked the specified question it identified the rejection at once, noting that “formally, you’re rejecting the joint restriction  $\gamma_{size} = \gamma_{B/M} = 0$ , which the CAPM requires,” and offered three interpretations: that the characteristics proxy for priced risks a single market index does not capture, following Fama and French; that “it’s the characteristics themselves, not the covariance structure, that predict returns,” following Daniel and Titman, which points to mispricing; and that a significant loading is “also consistent with the market proxy simply being wrong,” following Roll.

- (i) Yes. It identified the rejection immediately and stated the restriction the CAPM imposes.
- (ii) Yes, and more fully than required, contrasting the risk-based reading with the mispricing and invalid-proxy alternatives.
- (iii) Only partly. It noted that the characteristics are known before the return is earned, but did not make the point that lagging solves the econometric problem without settling whether the characteristics represent genuine risk factors.

Given our actual table it declined to claim a size effect and identified the beta slope as the main source of rejection, though it misstated  $t = 1.95$  as roughly 5% one-sided and 10% two-sided when the correct values are 2.6% and 5.2%. Asking the model to interpret real output proved a better test than asking it to explain a result stated in the prompt.

## 2 PART III, AI EXERCISE 2: PREDICTING CROSS-SECTIONAL DEVIATIONS

We asked an LLM which of the 49 industries it expected to fall above, below and near the security market line, and recorded its answer before comparing it with our results from Question (d).

### (a) The LLM’s predictions.

- *Above the line:* Tobacco, Utilities, Beer & Liquor / Food.
- *Below the line:* Coal / Mining, Computers / Electronic Equipment / Semiconductors, Autos / Steel.
- *Near the line:* Wholesale / Retail, Transportation, Financials / Business Services.

Its reasoning was the low-beta anomaly. Defensive industries have low beta but still earn competitive returns, so they plot above the line, while cyclical industries have high beta without the returns to match. It added a sin-stock premium for tobacco and a glamour effect for technology. It stated that these were predictions from documented patterns, not calculations from the data.

**(b) What the plot shows.** Deviations are measured from the theoretical SML in Figure 1, as  $\bar{R}_i - \bar{R}_M^e b_{iM}$ .

**Table 4:** Largest deviations from the theoretical security market line, balanced sample, percent per month. A positive value means the industry earned more than its beta implies.

| Above |        | Below |        | Near  |        |
|-------|--------|-------|--------|-------|--------|
| Smoke | +0.675 | Other | -0.468 | BldMt | -0.003 |
| Guns  | +0.461 | RlEst | -0.429 | Boxes | +0.004 |
| Soda  | +0.274 | PerSv | -0.376 | Telcm | +0.006 |

**(c) How accurate were the predictions?** The above-line predictions were accurate. All three industries plot above the line, and Tobacco is the largest positive deviation in the sample. The near-line predictions were also reasonable: Wholesale, Transportation and Financials all sit close to the line, though none is among the three closest.

The below-line predictions were mostly wrong. Coal and Mining plot slightly above the line rather than below it. Of the technology group, only Software is clearly below, while Chips and Electronic Equipment are above. Only Steel was placed correctly, and none of the three largest negative deviations was anticipated.

This asymmetry is explained by how the model reasoned. It used beta alone, predicting that low-beta industries plot above the line and high-beta industries plot below. Because the estimated line is so flat, the first half of this is almost automatic, since nearly any low-beta industry lands above the theoretical SML. That is why the above-line predictions worked. The below-line predictions needed more, because high beta is not sufficient: Chips and Electronic Equipment have high betas but earned enough to stay above the line. The industries that actually fall furthest below have betas near the sample average and unusually low returns, which reflects industry-specific weakness rather than a risk story the model could infer.

Its caveat about technology was partly right. It predicted technology would fall below the line but noted that recent data might reverse this. The reversal is what we observe for Chips and Electronic Equipment, though not for Software.